Project - Presentation

- A 10-15 minute presentation that will be presented to the class. The presentation should minally cover:
  - Overview of the project, AWS technologies used
  - A working demo running on AWS with a polished interface
  - No setup or teardown needed for demo
  - Presentation (minimally 8-10 slides) should be in PPT or PDF format
  - Everyone presents in person
- Presentations will be in class on **Mon 12/1** and **Wed 12/3**

Project - User's guide (Due 12/8)

- A user's guide should be submitted that includes:
  - Full description on how to **setup and use** your project
  - User's guide can be submitted either as a Word/PDF Document, md (Markdown) or HTML
  - Include examples, where relevant
  - There is no minimal length required
- Details on myCourses at Content > Project > 514 Term Project

Project - Submission Deliverables

- For each team, presentation, white paper and user's guide should be uploaded Assignments > Term Project > Project Submission
  - All code developed for this project, sample data (if any) and other relevant files must be uploaded to Github Classroom
  - For the presentations (proposal and final) and whitepaper, we will be using Google Docs so team progress and participation can be tracked
  - More details on myCourses at Content > Project > Term Project
- Grading Rubric:
  - Contains details for each deliverable of the project
  - See Content > Project > Term Project Grade Rubric

Setup Spark Cluster

- First - Do Assignments > Big Data > Setup EMR
  - There is a small charge for this activity
  - Stop after slide #3 (which you click "Create Cluster"), you are done for now (no screen shots needed)
  - This will take several minutes to run, and we will return to this shortly

RECAP

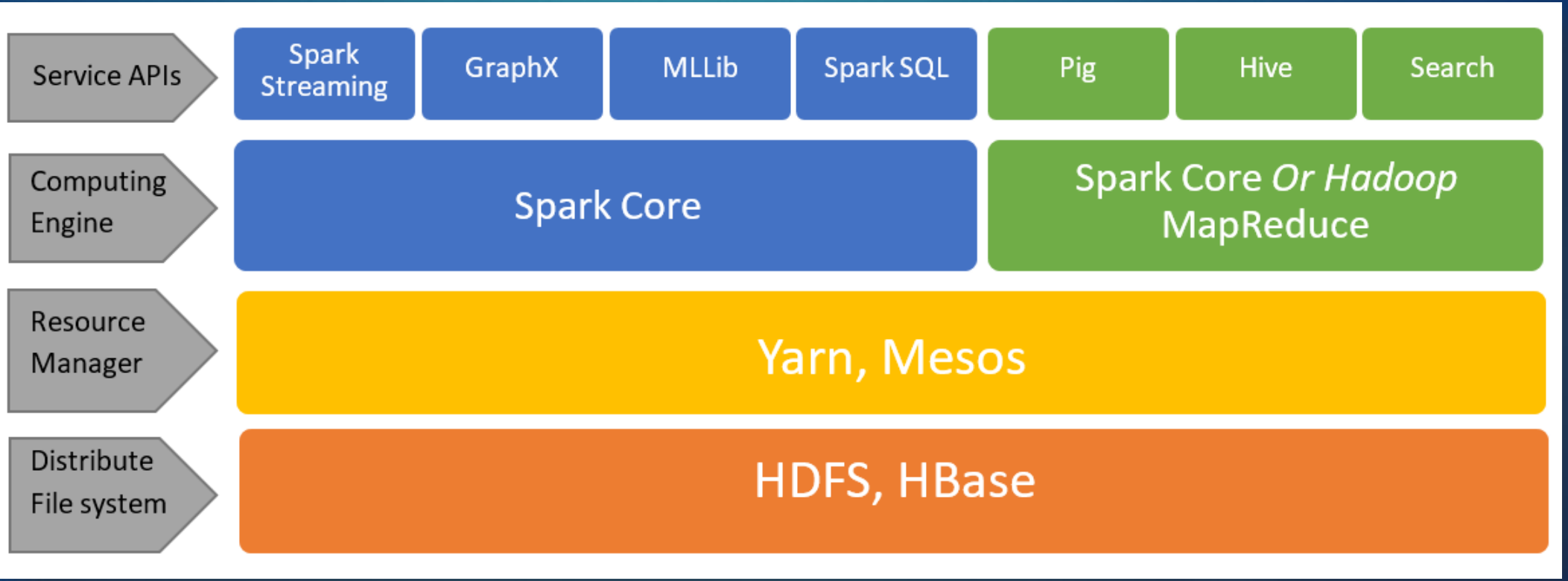
- What is Hadoop primarily designed to do?
  A. Store only structured relational data
  B. Provide large-scale storage and parallel processing on commodity servers ✅
  C. Replace all SQL-based databases
  D. Run only on specialized high-performance hardware

- In Spark, Transformations are executed immediately, while Actions are executed lazily
  - True
  - False ✅

- RDDs in Spark can be recomputed automatically if a node fails, making them fault-tolerant
  - True ✅
  - False

RECAP

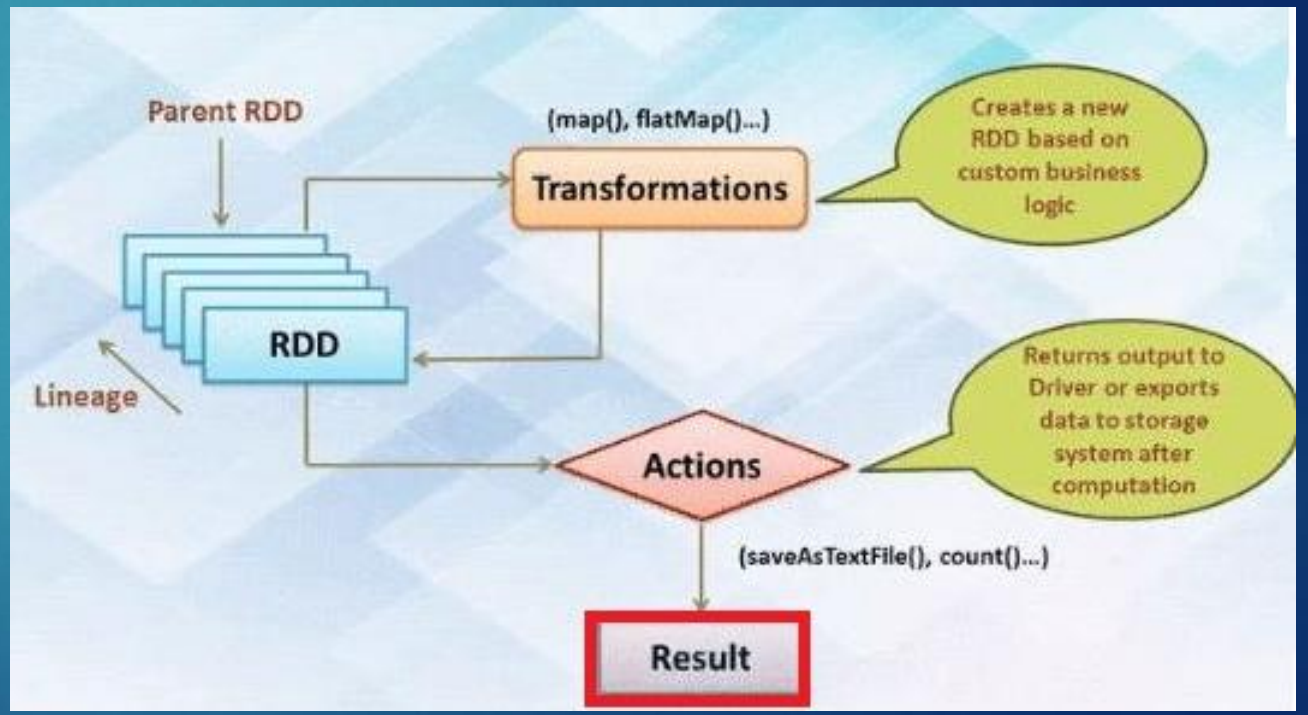
- Spark is an open-source, distributed processing system used for big data workloads
- The primary difference between Spark and Hadoop MapReduce is that Spark processes and retains data in memory for subsequent steps, whereas MapReduce processes data on disk



Source: https://towardsdatascience.com/apache-spark-101-3f961c89b8c

RECAP

- Spark has 2 types of RDD operation: Transformations and Actions
- Transformations are functions that take an RDD as the input and produce one or many RDDs as the output
- Actions are operations to access the actual data available in an RDD and provide non-RDD values
- Transformation's output is an input of Actions



Source: https://commandstech.com/spark-lazy-evaluation-with-example/

Big Data on the Cloud: Spark & Streaming

Engineering Cloud Software Systems

Department of Software Engineering
Rochester Institute of Technology

Big Data Streaming

- Big Data Streaming is a process in which large streams of real-time data are processed with the sole aim of extracting insights and useful trends out of it
- A continuous stream of unstructured data is sent for analysis into memory before storing it onto disk
- This happens across a cluster of servers and speed matters the most in big data streaming
- The value of data, if not processed quickly, decreases with time

Streaming vs. Batch

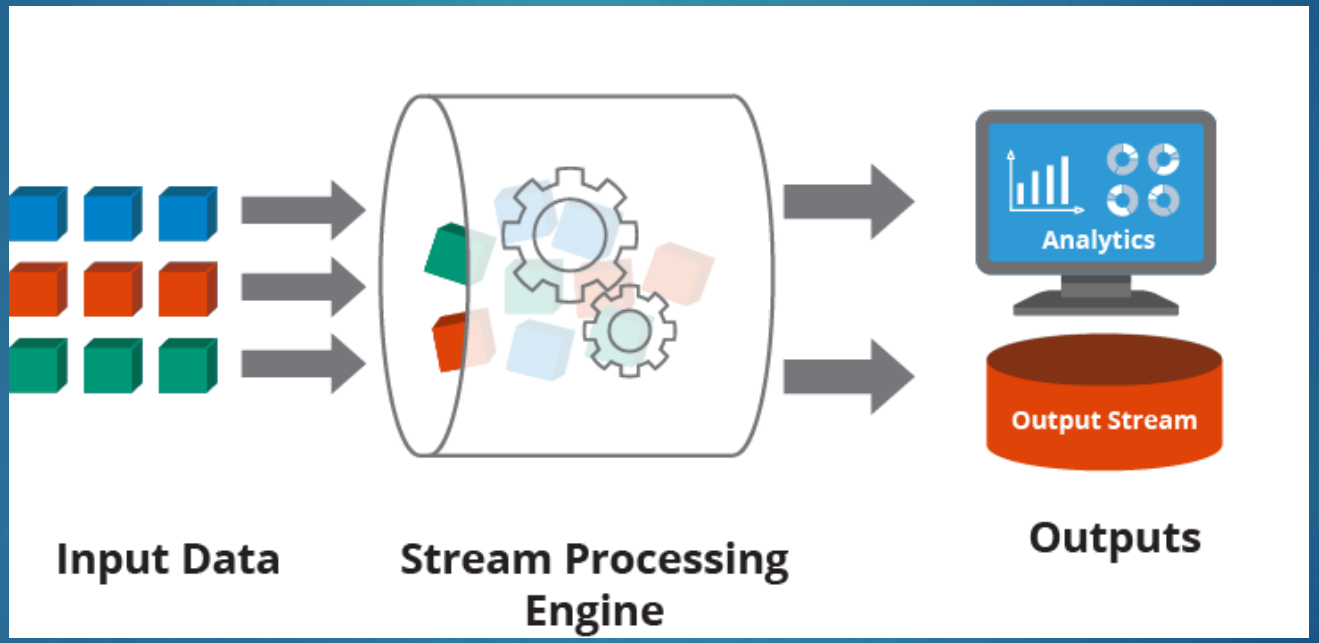
- Requires a set of data that is collected over time
- Handles a large batch of data
- Processes over all or most of the data
- Latency of the batch processing model will be in minutes to hours
- Lengthy process and is meant for large quantities of information that aren't time-sensitive

- Requires data to be fed into an analytics tool, often in micro-batches, and in real-time
- Handles individual records or micro-batches of few records
- Processes over data on a rolling window or most recent record
- Latency will be in seconds or milliseconds
- Processing is fast and is meant for information that is needed immediately

Source: https://k21academy.com/microsoft-azure/data-engineer/batch-processing-vs-stream-processing/

Streaming - Architecture

- Some examples of streaming data include IoT sensors, server and security logs, real-time advertising and click-stream data from apps and websites



- Data streams from one or more message brokers need to be aggregated, transformed and structured before data can be analyzed

Source: https://www.upsolver.com/blog/streaming-data-architecture-key-components

Examples of Streaming Data

- A **financial firm** streams stock market data to detect sudden changes, recalculate risk exposure, and auto-adjust client portfolios within seconds
- A **delivery company** tracks its vehicles in real time using GPS and sensor data. The system reroutes drivers instantly to avoid traffic, reduce fuel usage, and meet delivery windows
- **Wearable health devices** stream patient vitals (e.g., heart rate, oxygen levels) to care providers. Alerts are triggered if readings exceed safe ranges, enabling immediate medical intervention
- A **weather company** ingests satellite and ground sensor data in real time to issue severe weather alerts faster and help emergency teams respond proactively
- A **utility provider** gstreams power consumption data from smart meters. It adjusts rid load and usage dynamically, helping to prevent blackouts and optimize energy distribution

Source: https://aws.amazon.com/streaming-data/

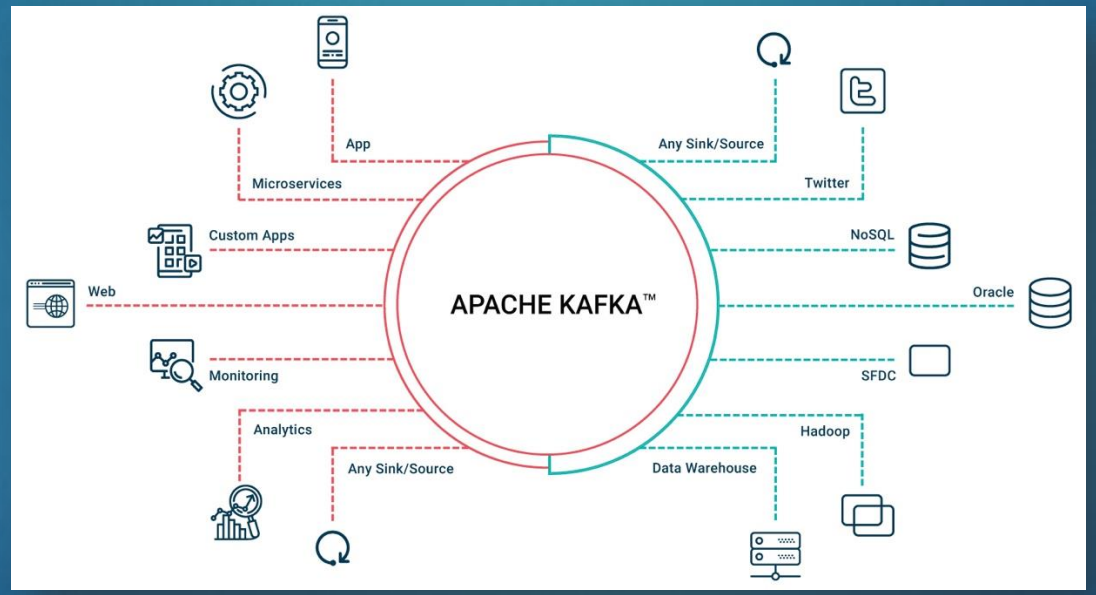
Streaming Technologies

- Apache Kafka and Amazon Kinesis are platforms for durable, reliable, and scalable data ingestion
- Both share common concepts: producers, consumers, partitioning/sharding, and replication
- Designed for real-time processing, they can ingest thousands of simultaneous data feeds to support high-speed analytics and applications

Source: https://www.spec-india.com/blog/kinesis-vs-kafka

Kafka

- Apache Kafka (originally development at LinkedIn) is a distributed event store and stream-processing platform
- Its core architectural concept is an immutable log of messages that can be organized into topics for consumption by multiple users or applications



- It can be deployed on bare-metal hardware, virtual machines and containers (on-premise and cloud)

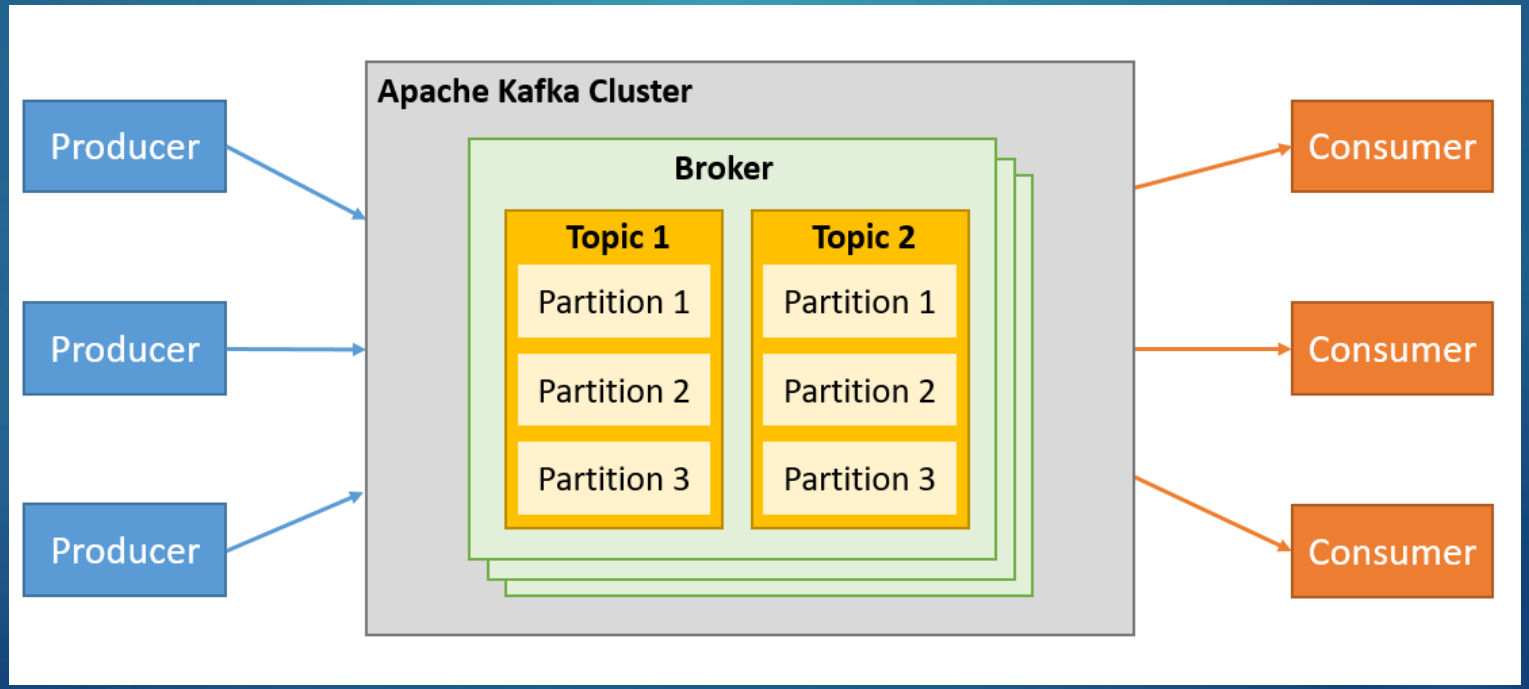
Kafka

- Kafka is used by thousands of companies including over 80% of the Fortune 100

Source: https://soshace.com/the-ultimate-introduction-to-kafka-with-javascript/

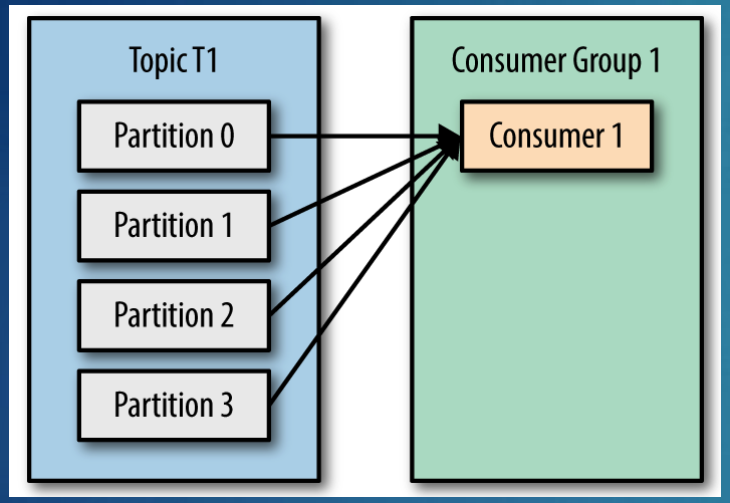
Kafka

- Kafka stores messages in topics that are partitioned and replicated across multiple brokers in a cluster
- Producers send messages to topics from which consumers read
  - Within a topic, partitions can be added to improve scalability
- Consumers read messages from partitions within a topic



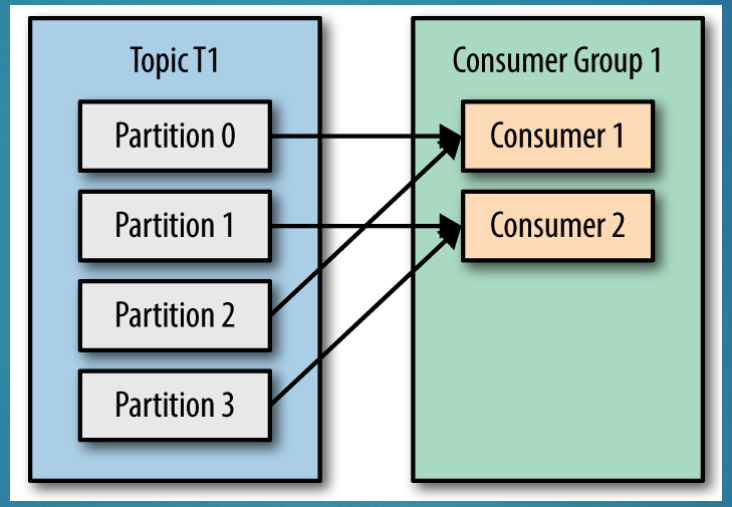
Kafka – Consumer and Consumer Groups

- Kafka Consumers are part of a Consumer group
- Question: What is a "Consumer" an example of?

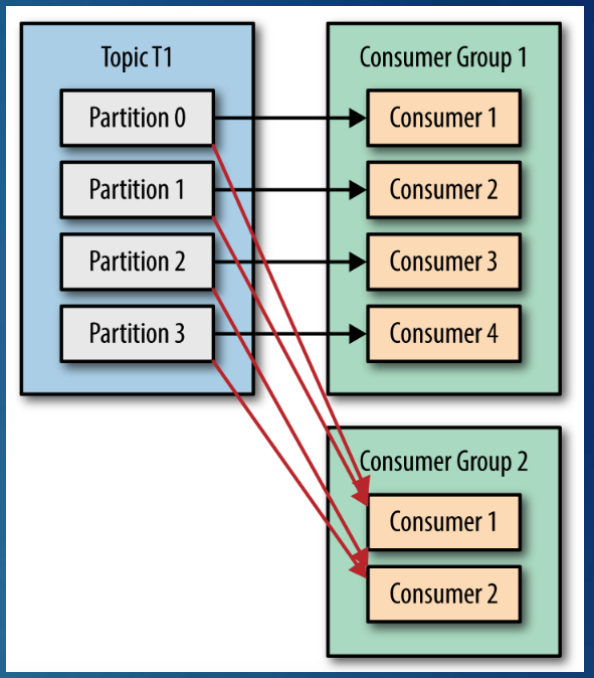


Consumer 1 will get all messages from all four T1 partitions

What happens if Consumer 1 cannot keep up with all the partitions?



Adding another Consumer to the group allows the 2 Consumers to consume the messages from the partitions more efficiently

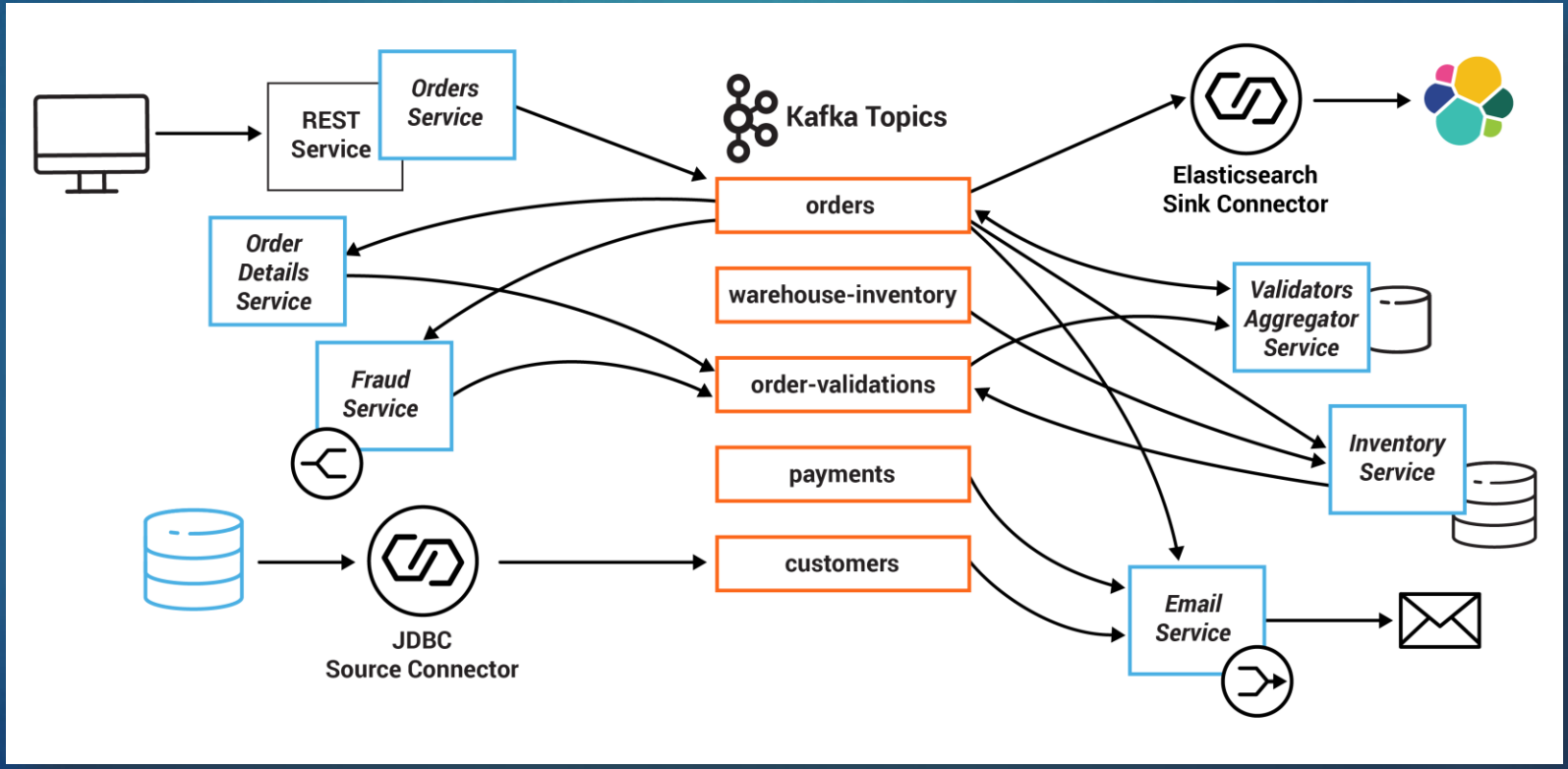


Two Consumer groups can independently process messages from the T1 partitions

Source: https://www.oreilly.com/library/view/kafka-the-definitive/9781491936153/ch04.html

Kafka Microservices Architecture - Example

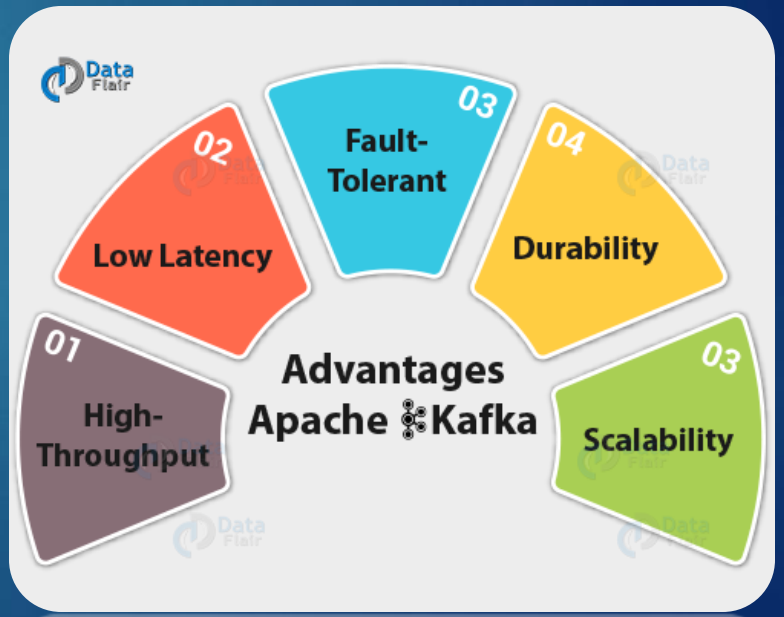
- A system centers on an Orders Service which exposes a REST interface to POST and GET Orders
- Posting an Order creates an event in Kafka that is recorded in the topic orders
- This is picked up by different validation engines (Fraud Service, Inventory Service and Order Details Service), which validate the order in parallel, emitting a PASS or FAIL based on whether each validation succeeds



Source: https://docs.confluent.io/platform/current/tutorials/examples/microservices-orders/docs/index.html

Kafka – Benefits

- Scalable Architecture
  - Kafka's partition-based design distributes data across multiple brokers, enabling it to handle massive volumes of streaming data that go far beyond a single server's capacity
- High Performance & Low Latency
  - Kafka decouples producers and consumers, enabling real-time data flow with minimal delay, making it ideal for high-throughput systems
- Durable & Fault-Tolerant
  - Kafka is designed to withstand server failures through built-in replication and persistent storage, ensuring that data remains safe and available
- Reliable & Flexible Performance
  - Kafka offers strong durability guarantees, scalable pub/sub and queue models, configurable consistency, and message ordering within partitions—all of which support stable, high-throughput applications

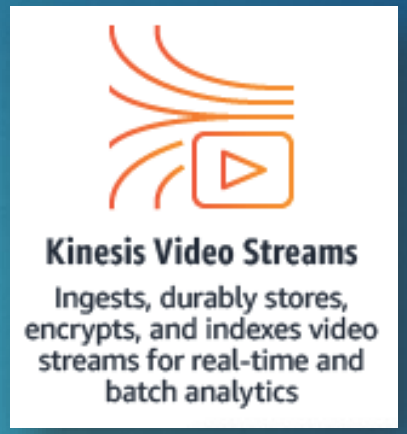


Source: https://scalac.io/blog/what-is-apache-kafka-and-what-are-kafka-use-cases/

Kinesis

- Amazon Kinesis is a fully managed service that collects, processes, and analyzes real-time streaming data at any scale, enabling instant insights and actions from sources like IoT devices, clickstreams, and video feeds
- Key capabilities:
  - **Real-Time Streaming** – Process and analyze data as it arrives for immediate insights
  - **Scalable & Flexible** – Seamlessly handle any volume of data with pay-as-you-go pricing
  - **Faster Decisions** – Eliminate batch delays, accelerating automation and decision-making
  - **Diverse Sources** – Stream from video, audio, IoT sensors, application logs, and clickstreams
  - **Advanced Use Cases** – Power machine learning, anomaly detection, predictive analytics, and dashboards

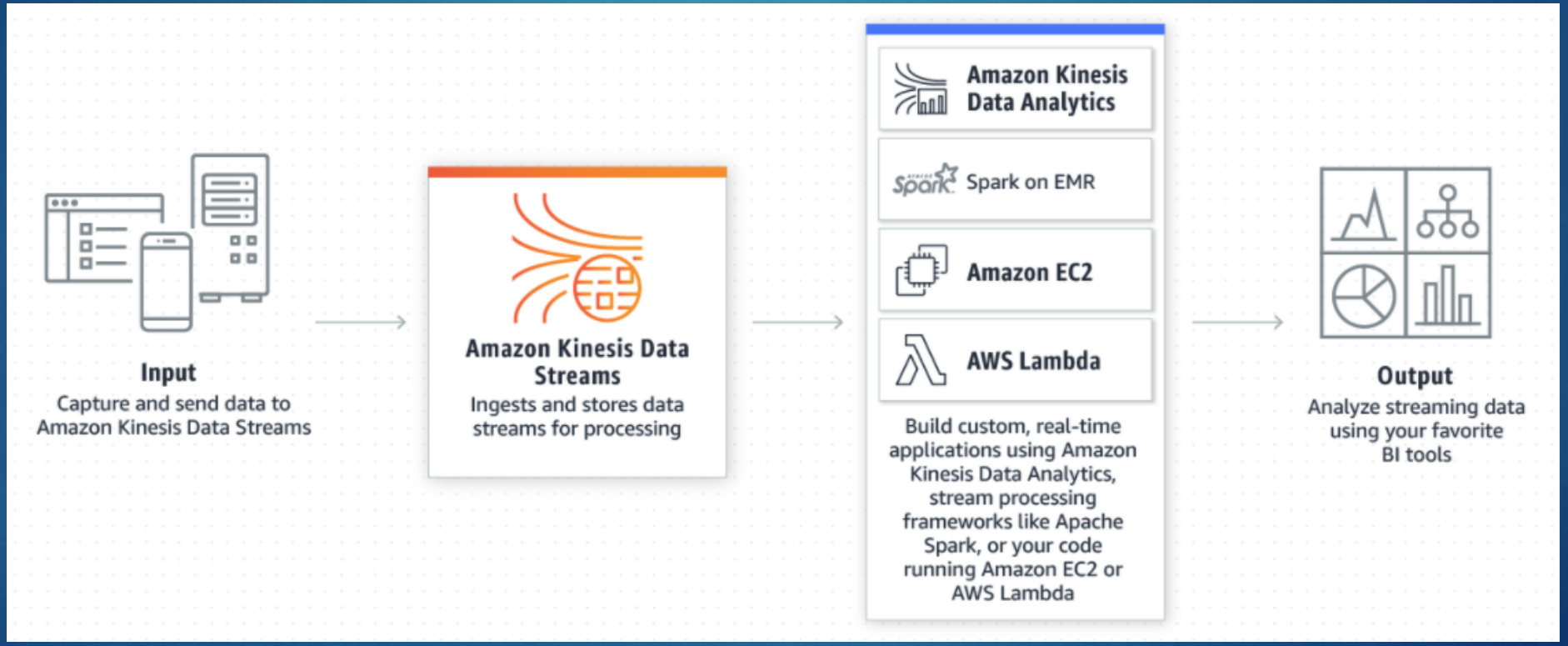
Kinesis Capabilities



- Real-time
  - Enables ingesting data in real-time, so you can run analytics queries instantly without waiting for your data to accumulate to analyze in future
- Fully managed
  - You do not have to maintain servers or run any software upgrades
- Scalable
  - Automatically scales up and down and can handle any amount of incoming data

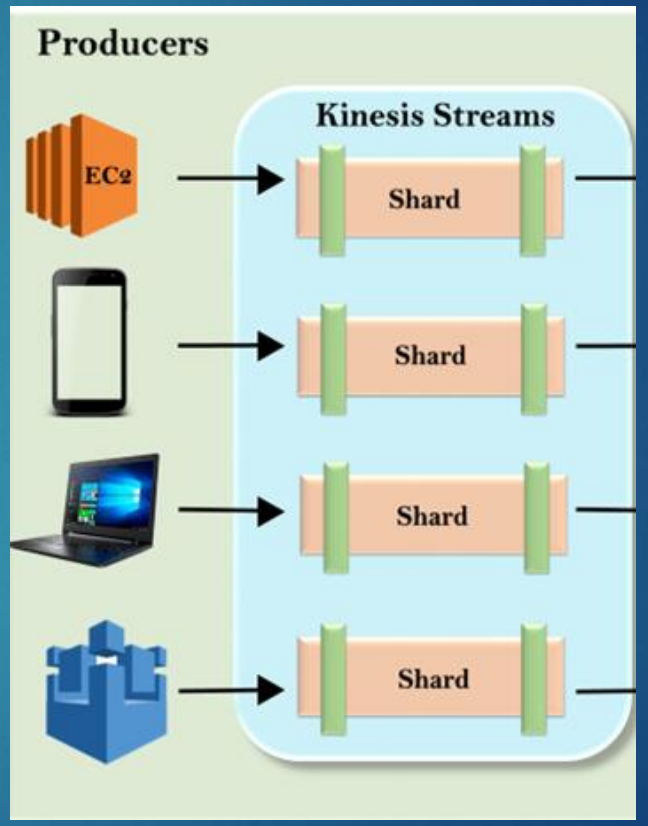
Kinesis Data Streams

- Data Producers enter the records into Kinesis Data Streams (KDS)
- AWS offers the Kinesis Producer Library for simplifying producer application development



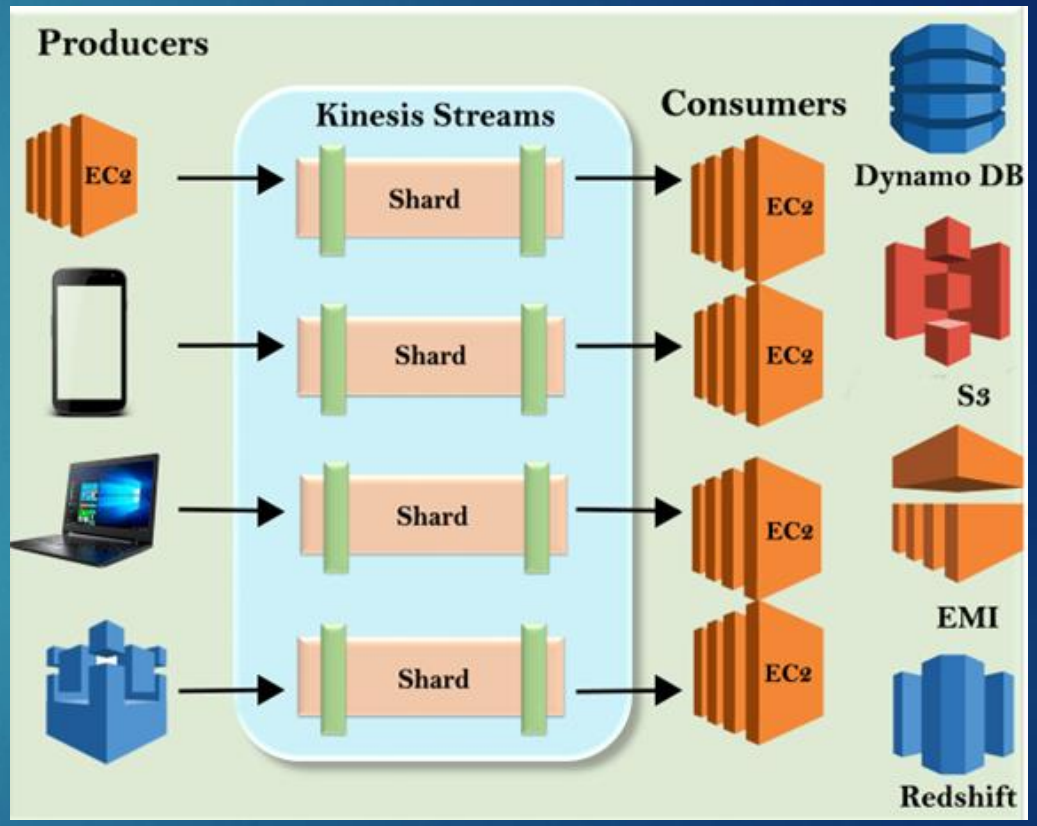
Kinesis Data Streams - Producers

- Producers write records into a Kinesis Data Stream, which is composed of one or more shards.
- Each shard provides:
  - Writes: Up to 1,000 records per second (maximum 1 MB/sec)
  - Reads: Up to 5 transactions per second (maximum 2 MB/sec)
  - The capacity of a stream is determined by the number of shards you configure
  - The total throughput of the stream equals the combined capacity of all shards



Kinesis Data Streams - Consumers

- Consumers are applications that process data records from a Kinesis Data Stream
- Each consumer receives its own 2 MB/sec read throughput, independent of other consumers
- Multiple consumers can read from the same stream in parallel, without competing for read capacity



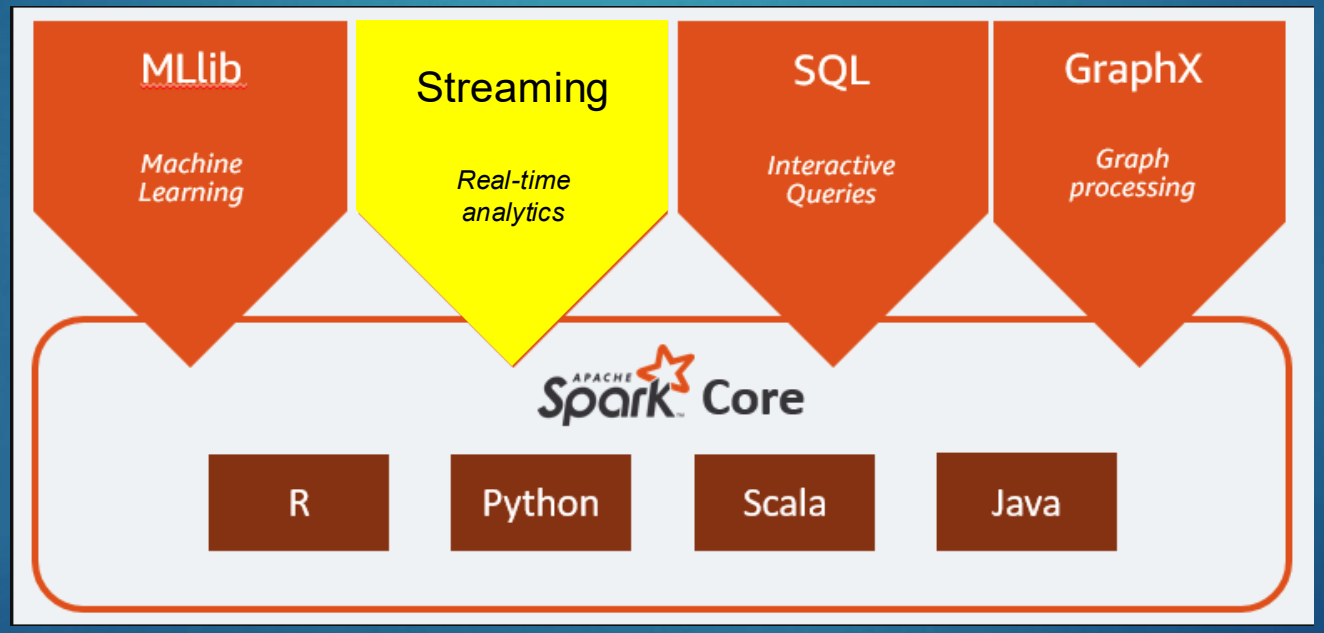
Kinesis or Kafka?

Service	Strengths	Challenges / Limitations	Best For
Apache Kafka	• Full flexibility & feature set • Latest version support • Highly customizable	• Requires significant setup & ongoing management • Higher DevOps overhead	Teams with strong in-house expertise needing maximum control
Amazon Kinesis	• Fully managed AWS service • Simple setup & integration • Scales easily with no infra management	• More limitations than Kafka • Less flexible for custom use cases	Teams that want **easy streaming** without managing infrastructure
Amazon MSK (Managed Streaming for Apache Kafka)	• Managed Kafka on AWS • Reduces operational burden • Uses native Kafka APIs	• Not always fully up to date with latest Kafka releases • Less flexible than self-managed Kafka	Teams that want **Kafka features** but prefer AWS to handle infrastructure

- In summary:
  - Kafka ➔ Control
  - Kinesis ➔ Simplicity
  - MSK ➔ Middle ground

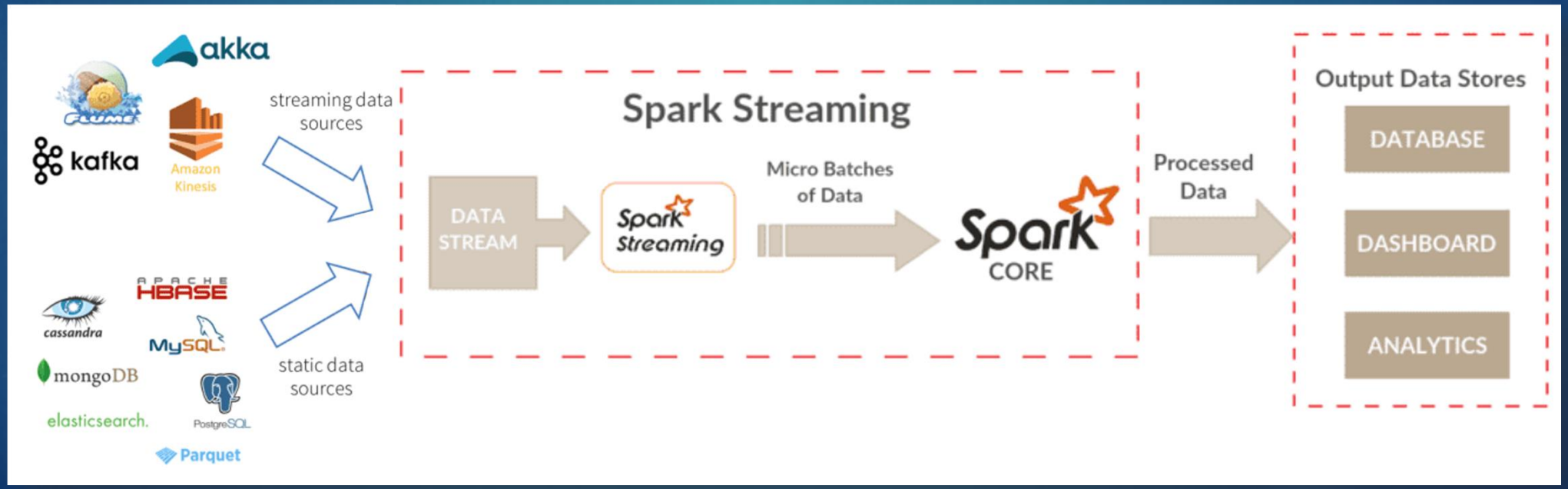
Spark Streaming

- Spark Streaming is an extension of the core Spark API that enables scalable, high-throughput, fault-tolerant stream processing of live data streams



Spark Streaming & Data Sources

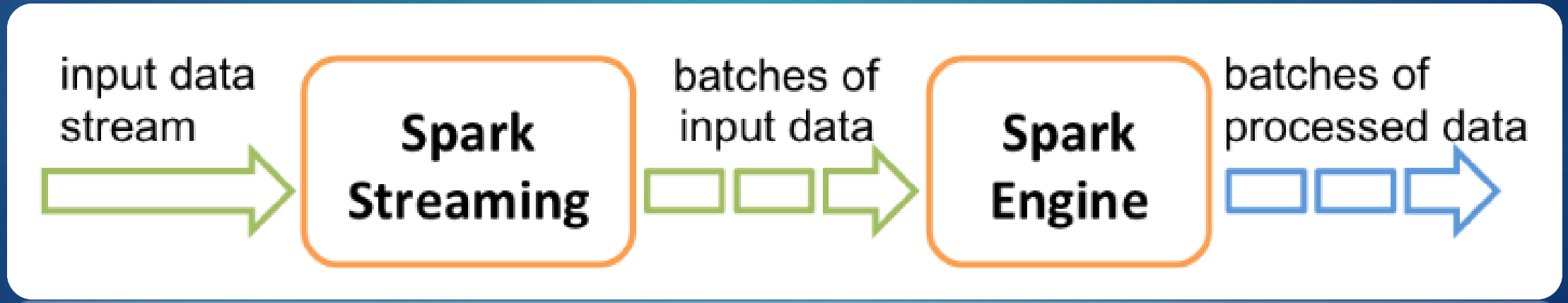
- Spark Streaming is an extension of the Spark API that enables real-time data processing for data engineers and data scientists
- It can ingest data from a wide range of sources, including Apache Kafka, Amazon Kinesis, and other streaming platforms
- The processed data can then be delivered to file systems, databases, or live dashboards to provide immediate insights



Source: https://www.databricks.com/glossary/what-is-spark-streaming

Spark Streaming – How does it work?

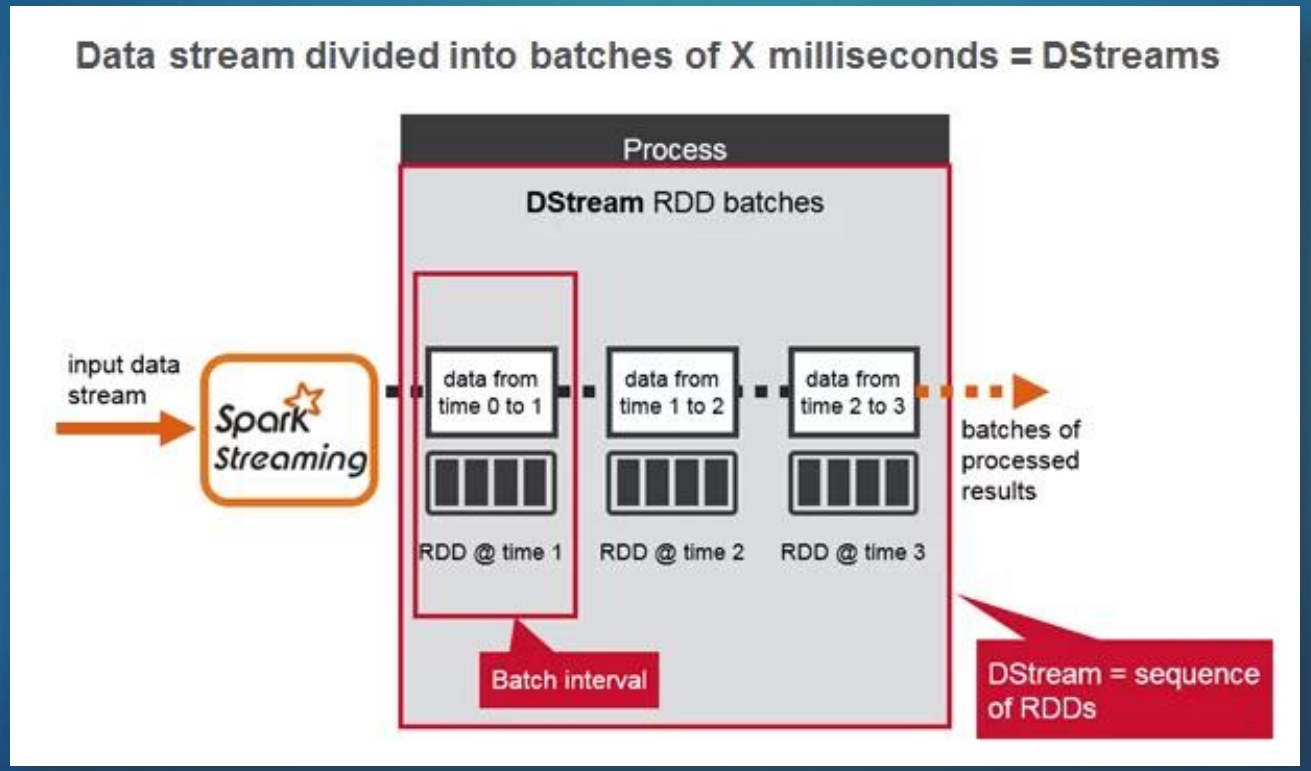
- **Real-Time Processing** – Ingests live data streams and processes them in small batches using the Spark engine
- **DStreams (Discretized Streams)** – A high-level abstraction that represents a continuous stream of data
- **RDD-based Architecture** – Each DStream is internally represented as a sequence of RDDs, enabling parallel and fault-tolerant processing



Source: https://spark.apache.org/docs/latest/streaming-programming-guide.html#overview

Spark Streaming – How does it work?

- Incoming data is grouped into batch intervals (based on time)
- Each batch interval generates a new RDD
- Every RDD contains the records collected during that specific interval



Source: https://www.javacodegeeks.com/2016/04/fast-scalable-streaming-applications-mapr-streams-spark-streaming-mapr-db.html

Spark Streaming – Simple example

- Wordcount.py - Streaming style

```
from pyspark import SparkContext
from pyspark.streaming import StreamingContext

sc = SparkContext("local[2]", "NetworkWordCount")
ssc = StreamingContext(sc, 1)

lines = ssc.socketTextStream("localhost", 9999)

wordCounts = lines.flatMap(lambda x: x.split(' ')) \
        .map(lambda x: (x, 1)) \
        .reduceByKey(lambda x,y: x + y)

wordCounts.pprint()

ssc.start()             # Start the computation
ssc.awaitTermination()  # Wait for the computation to terminate
```

Create a local StreamingContext and batch interval of 1 second

*Create a **DStream** that will connect to hostname:port, like localhost:9999*

This is the same code from our previous wordcount example using an RDD

Print the first 10 elements of the RDD

Loops waiting for events

- In a separate window, run Netcat (small utility found on most Linux/Unix systems)

```
nc -lk 9999
```

Writing to port 9999

Source: https://spark.apache.org/docs/latest/streaming-programming-guide.html#overview

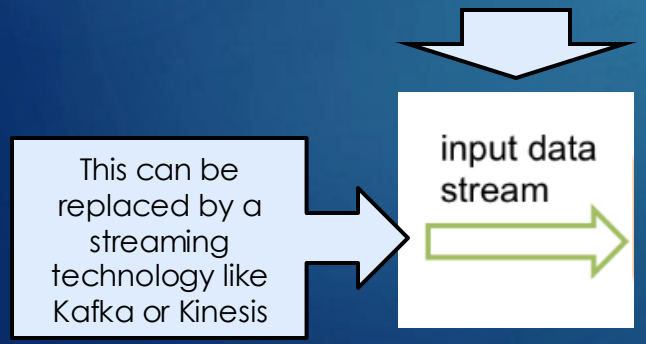
Spark Streaming – Simple example

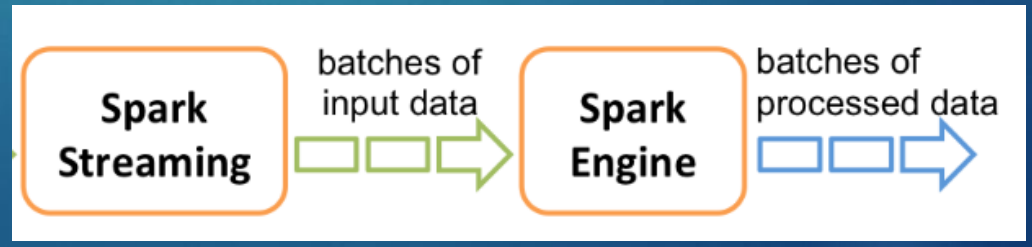
Window #1 running Netcat

```
# TERMINAL 1:
# Running Netcat

$ nc -lk 9999

hello world
```

Window #2 running WordCount.py

```
$ spark-submit Wordcount.py localhost 9999
...
-------------------------------------------
Time: 2014-10-14 15:25:21
-------------------------------------------
(hello,1)
(world,1)
```

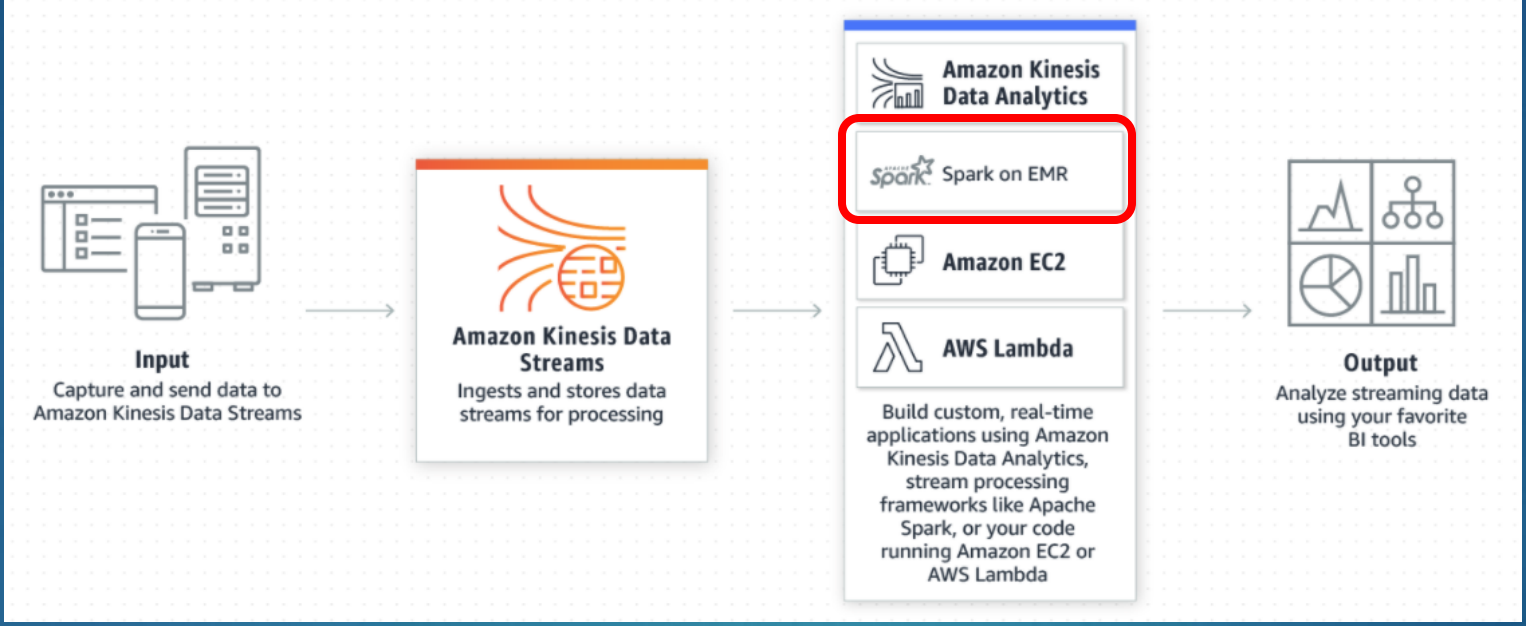




Source: https://spark.apache.org/docs/latest/streaming-programming-guide.html#overview

Kinesis Data Streams

- With Spark on EMR, you can create a DStream that connects to a Kinesis Data Stream, which can process the data as RDDs

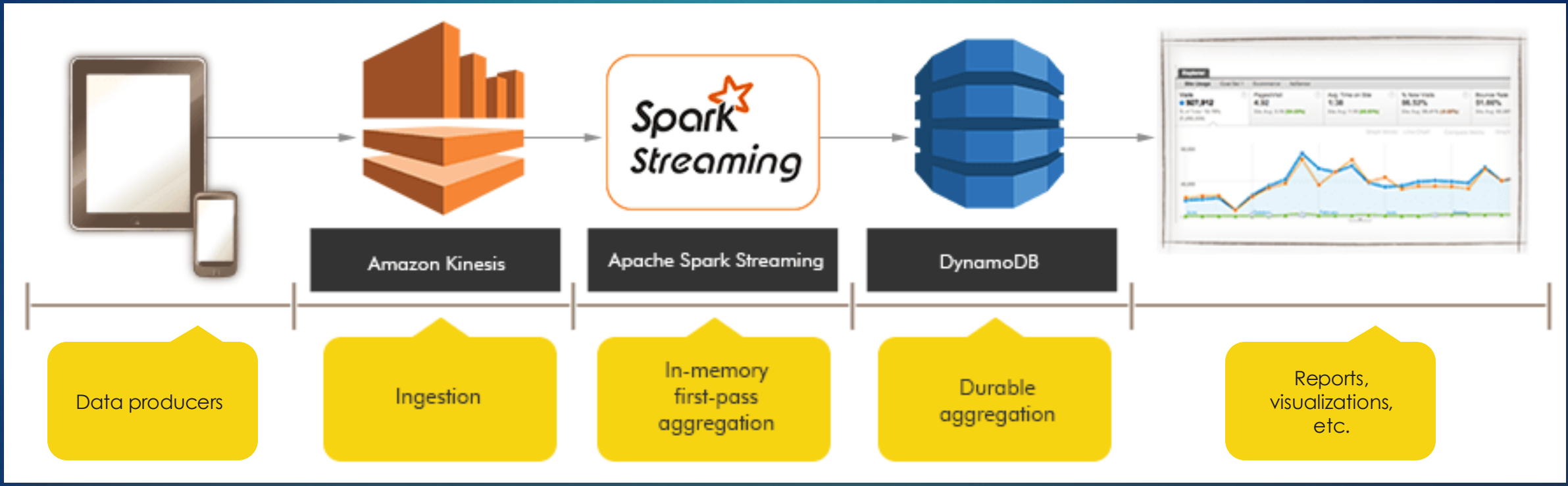


Creates a **DStream** from a Kinesis Data Stream

```
from pyspark.streaming.kinesis import KinesisUtils, InitialPositionInStream

kinesisStream = KinesisUtils.createStream( streamingContext, [Kinesis app name],
[Kinesis stream name], [endpoint URL], [region name], [initial position],
[checkpoint interval], StorageLevel.MEMORY_AND_DISK_2)
```

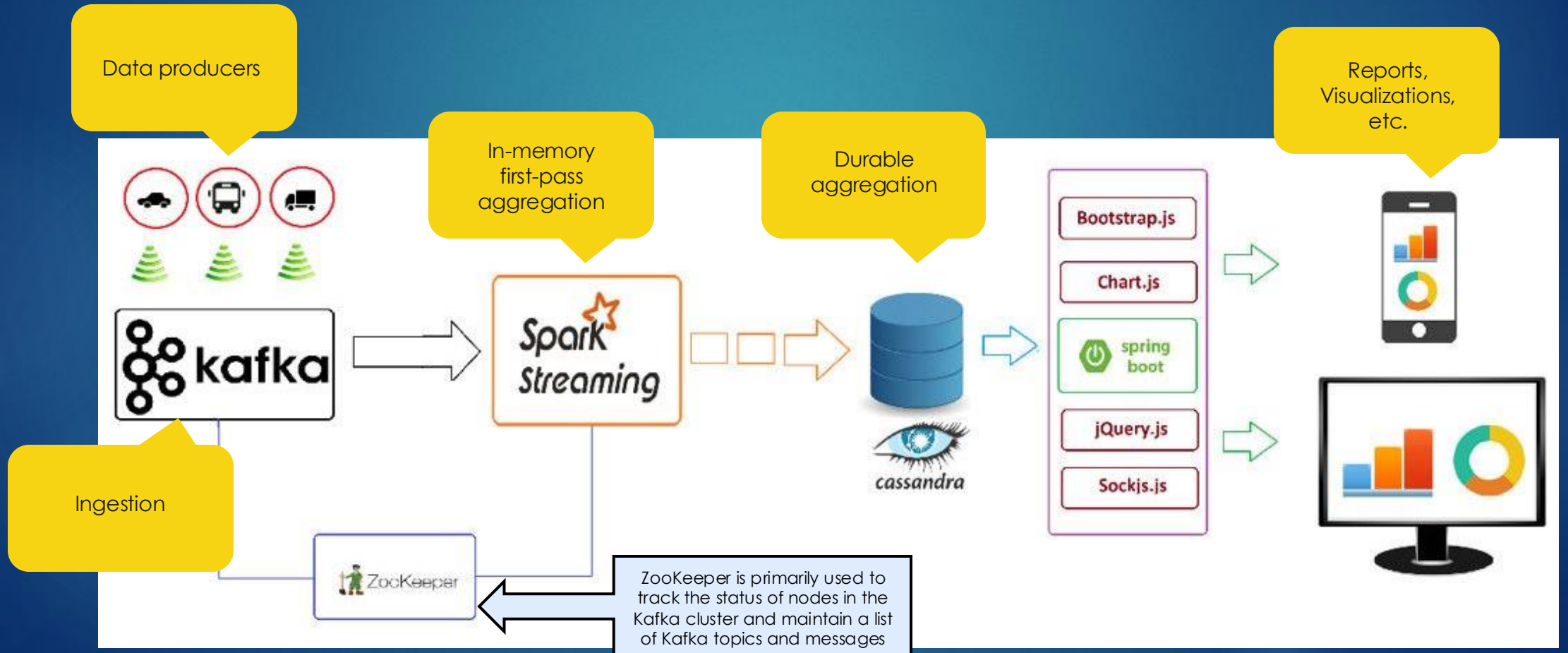
Kinesis + Spark Streaming

- Combining this all together



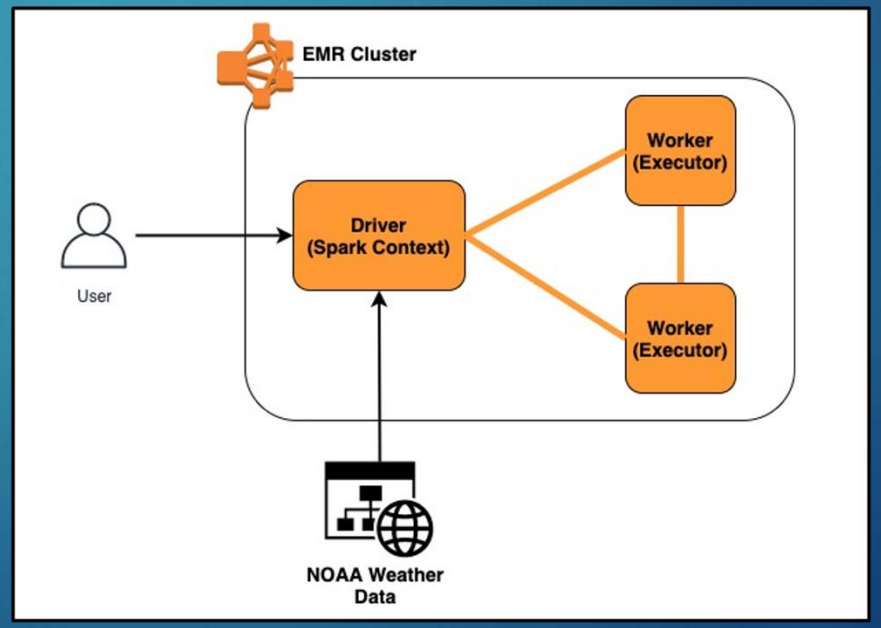
Kafka + Spark Streaming

- Combining this all together



Weather modelling - Exercise

- This exercise goes beyond Wordcount and provides a more realistic use case using Spark
- We will use data from NOAA (National Oceanic and Atmospheric Administration)
  - This data is from Western-European weather stations from 1980 to 2014
- Follow the instructions in Assignments > Activity - Evaluating Weather Data with Spark
  - Part 1 due by end of class



- There are 3 deliverables plus an opportunity for a **bonus point** in this activity